

Assessment – 3

Dry Run Test

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Question 1: Dry Run of A Search Algorithm*

(A)

/ \

2/ \4

/ \

(B)----3----(C)

| \ |

7 | \2 |3

| \ |

(D)----2----(E)

\

\2

(G)

Node h(n)

A 7

B 6

C 4

D 3

E 2

G 0

Task:

Perform a **dry run of the A*** Search algorithm from **A to G**.

For **each iteration**, show:

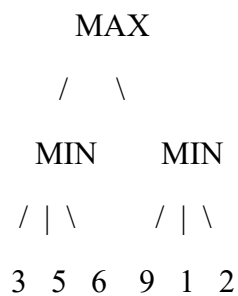
- Current node
- Open List
- Closed List
- $g(n)$
- $h(n)$
- $f(n) = g(n) + h(n)$

Finally:

- Identify the **optimal path**
- Calculate the **total path cost**

Question 2: Dry Run of Minimax with Alpha-Beta Pruning

Consider the following game tree:



Task:

Perform a **dry run of the Minimax algorithm with Alpha-Beta Pruning**, evaluating the tree **from left to right**.

For **each node**, show:

- Alpha (α)
- Beta (β)
- Selected value
- Any node that gets **pruned**

Finally:

- Determine the **best move for MAX**

- State the **final minimax value**
- List the **pruned nodes (if any)**